

# Structural Infinite-Exponent Partition Relations and Weak Choice Principles

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We investigate infinite-exponent partition relations on arbitrary relational structures, with a focus on linear orders and graphs. Any such relation contradicts the Axiom of Choice. We show that there are some such relations which are consistent with ZF which imply the failure not just of Choice but also of the Kinna-Wagner Selection Principle  $KWP_1$  and the Ordering Principle  $O$ .

## 1 Introduction

### 1.1 Background and motivation

The results in this paper are an offshoot of the work done in [GSW26]. Several of our results in that paper show in ZF that for certain order types  $\tau$ , a partition relation with exponent  $\tau$  can never hold on an order of the form  $\langle {}^\alpha 2, <_{\text{lex}} \rangle$ , for  $\alpha$  an ordinal, and by extension can never hold on any linear order  $\langle L, < \rangle$  which embeds into some  $\langle {}^\alpha 2, <_{\text{lex}} \rangle$  (partition relations and the arrow notation used to express them will be explained in §1.2). In ZFC, every linear order  $\langle L, < \rangle$  embeds into some  $\langle {}^\alpha 2, <_{\text{lex}} \rangle$ , but this is not in general true in ZF. It is natural, then, to ask whether there consistently exists *any*  $\langle L, < \rangle$  for which

$$\langle L, < \rangle \rightarrow (\tau)^\tau.$$

The results of Section 2 show that in a number of cases the answer is yes. The particular linear orders  $\langle L, < \rangle$  we obtained had the property that the elements of  $L$  were sets of sets of ordinals, leading naturally to the question of whether this was necessary, or whether there consistently existed such  $\langle L, < \rangle$  with  $L \subseteq {}^\alpha 2$  for some  $\alpha$ . In Section 3 we show that it is in fact necessary, and that any such  $\langle L, < \rangle$  has the property that the set  $L$  does not inject into the power set of an

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ordinal, a failure of the Kinna-Wagner Selection Principle. Similar results in the setting of graphs are also proved in both sections.

## 1.2 Partition relations

For  $A, B$  structures in some relational language, write  $[A]^B$  to denote the set of substructures of  $A$  which are isomorphic to  $B$  (the set of *copies of  $B$  in  $A$* ).<sup>1</sup> For  $A, B, C$  structures in the same language with  $[A]^B$  and  $[B]^C$  non-empty and  $\chi$  a set, the *partition relation*

$$A \rightarrow (B)_\chi^C$$

is the statement that for any  $F : [A]^C \rightarrow \chi$ , thought of as a *colouring* of the copies of  $C$  in  $A$ , there is some  $H \in [A]^B$  which is *homogeneous* (or *monochromatic*) for  $F$ , in the sense that  $|F \restriction [H]^C| = 1$ . When  $\chi$  is omitted it is understood to equal  $2 = \{0, 1\}$ . The relation  $A \rightarrow (B)_\chi^C$  is clearly preserved by replacing any of  $A, B, C$  with isomorphic structures; in the particular setting of linear orders, we will often use *order types*, defined in §1.3 in place of concrete orders in this notation. The partition relation has monotonicity in its terms in the following sense: the relation  $A \rightarrow (B)_\chi^C$  is preserved if  $A$  is replaced by a *larger* structure  $A'$ , if  $B$  is replaced by a *smaller* structure  $B'$ , or if  $\chi$  is replaced by a smaller set  $\chi'$ .<sup>2</sup> We do not in general have monotonicity in the *exponent*  $C$ .

Our work focuses on *infinite-exponent partition relations* (IEPRs), and in particular on relations whose exponent  $C$  contains proper subcopies of itself: in this setting, even the *minimal relation*  $A \rightarrow (C)^C$  is not guaranteed to hold. Indeed, under the Axiom of Choice, such a relation can never hold; it is in light of this that we work in ZF without Choice when discussing IEPRs.

**Proposition 1.** (ZFC) *Let  $A$  and  $C$  be structures such that  $[A]^C \neq \emptyset$  and  $[C]^C \not\supseteq \{C\}$ . Then*

$$A \not\rightarrow (C)^C.$$

*Proof.* Under AC, every set is well-orderable; fix an enumeration  $\langle X_\alpha : \alpha < \gamma \rangle$  of  $[A]^C$  for some  $\gamma$ . Say that  $\alpha$  is the *index* of  $X_\alpha$ . We inductively define a colouring  $F : [A]^C \rightarrow 2$  with no homogeneous set.

For  $\alpha < \gamma$ , at stage  $\alpha$ , if the value of  $F(X_\alpha)$  has already been determined, proceed to stage  $\alpha + 1$ ; otherwise, determine the value of  $F(X_\alpha)$  by means of the following procedure. If there exists a descending  $\omega$ -chain  $X_\alpha = X_{\alpha_0} \supsetneq X_{\alpha_1} \supsetneq X_{\alpha_2} \supsetneq \dots$  such that  $F(X_{\alpha_n})$  has not been decided for any  $n \in \omega$ , choose such a chain, and for each  $n$  set

$$F(X_{\alpha_n}) := \begin{cases} 0 & \text{if } n \text{ is even;} \\ 1 & \text{if } n \text{ is odd.} \end{cases}$$

<sup>1</sup>In structural Ramsey theory, this is usually denoted  $\binom{A}{B}$ , but we avoid this notation as it clashes with the notation used for polarised partition relations (cf. [Ga25, §4], [GSW26, §4]).

<sup>2</sup>More precisely, we mean  $A', B', \chi'$  such that  $[A']^{A'} \neq \emptyset$ ,  $[B]^{B'} \neq \emptyset$ , and  $\chi'$  injects into  $\chi$ .

In this case say that the  $X_{\alpha_n}$  have been *coloured by alternation*. If it is not possible to find such a sequence, simply set  $F(X_\alpha) = 0$ .

It is clear that any  $X \in [A]^C$  which was coloured by alternation is not homogeneous for  $F$ , and also that if any  $X_\alpha$  was coloured at a stage  $\beta < \alpha$ , it was coloured by alternation. We claim that no  $X \in [A]^C$  is homogeneous for  $F$ ; to see this, it suffices to show that for every  $X \in [A]^C$ , there is some  $X' \in [X]^C$  which was coloured by alternation. Let  $X = X_\alpha$  be arbitrary. Let  $\alpha'$  be the minimal index of an element of  $[X_\alpha]^C$ , and consider  $X_{\alpha'} \in [X_\alpha]^C$ . Since  $\alpha'$  is the minimal index of an element of  $[X_\alpha]^C$ , it is also the minimal index of an element of  $[X_{\alpha'}]^C$ , i.e. for any  $X_\beta \in [X_{\alpha'}]^C$ ,  $\beta \geq \alpha'$ . If  $X_{\alpha'}$  was coloured by alternation, we are done; otherwise, it must be the case that at stage  $\alpha'$ , some of the elements of  $[X_{\alpha'}]^C$  already had colours assigned, such that it was impossible to find a descending chain of substructures whose colour under  $F$  had not yet been determined. But for  $F(X_\beta)$  to have been determined by stage  $\alpha' < \beta$ , it must have been coloured by alternation.  $\square$

It is therefore the case that any IEPR of the form  $A \rightarrow (C)^C$  implies the failure of AC; in Section 3, we show that certain IEPRs which are consistent with ZF also imply the failure of weaker choice principles than the full Axiom of Choice.

### 1.3 Linear orders

Much of this paper relates to the combinatorics of linear orders. We reserve the letters  $\sigma$  and  $\tau$  for *order types*, i.e. isomorphism classes of linear orders under order-isomorphism. Addition of linear orders or order types refers to concatenation, so e.g.  $\sigma + \tau$  is the order type of a linear order consisting of an initial segment ordered as  $\sigma$  followed by a final segment ordered as  $\tau$ . Multiplication is colexicographic, so  $\sigma\tau$  refers to the order type of  $\tau$ -many copies of  $\sigma$ . For  $\sigma$  an order type,  $\sigma^*$  denotes its reverse.  $\omega$  is the order type of the natural numbers, and  $\eta$  is the order type of the rational numbers, each under their usual ordering. The *embeddability relation*  $\leq$  is defined as follows:  $\sigma \leq \tau$  iff any  $\langle L, < \rangle$  of type  $\tau$  has a suborder of type  $\sigma$ .

For  $\langle L, < \rangle$  a linear ordering, a *condensation class* of  $\langle L, < \rangle$  (under the finite condensation) is an equivalence class of the relation  $x \sim_L y$  given by

$$x \sim_L y \iff \{z \in L : x \leq z \leq y \text{ or } y \leq z \leq x\} \text{ is finite.}$$

Condensation classes are necessarily either finite or they are ordered as one of  $\omega$ ,  $\omega^*$ , or  $\omega^* + \omega$ . We will be particularly interested in the condensation classes of certain subsets  $A \subseteq L$  of an ambient linear order  $\langle L, < \rangle$ , equipped with the induced suborder. For  $A \subseteq L$ , write  $\mathbf{cc}^\omega(A)$  for the set of condensation classes of  $A$  ordered as  $\omega$ , and  $\mathbf{cc}^{\omega^*}(A)$  for the set of condensation classes of  $A$  ordered as  $\omega^*$ .

A linear order  $\langle L, < \rangle$  is said to be *scattered* if no suborder of  $\langle L, < \rangle$  is densely ordered; equivalently, for  $L$  well-orderable,  $\langle L, < \rangle$  is scattered iff  $[\langle L, < \rangle]^\eta = \emptyset$ .

Iterations of the finite condensation allow for a detailed analysis of the structure of scattered orders; see [Ro82, §5].

Intervals are written with parentheses and brackets as usual;  $(x, \rightarrow)$  and  $[x, \rightarrow)$  denote final segments, and  $(\leftarrow, x)$ ,  $(\leftarrow, x]$  denote initial segments. Some of our proofs involve sets equipped with multiple different linear orderings; in these settings, we write e.g.  $(x, y)_{<}$  and  $(x, y)_{<_{\text{lex}}}$  to disambiguate which order is meant when speaking of intervals.

An order type  $\sigma$  is (*additively*) *indecomposable* if whenever  $\sigma$  is written as a sum  $\sigma = \tau + \varphi$ , either  $\sigma \leq \tau$  or  $\sigma \leq \varphi$ . Further,  $\sigma$  is *strictly indecomposable to the left* (resp. *right*) if whenever  $\sigma$  is written as a sum  $\sigma = \tau + \varphi$ ,  $\sigma \leq \tau$  (resp.  $\sigma \leq \varphi$ ).

The orders  $\langle {}^\alpha 2, <_{\text{lex}} \rangle$  for  $\alpha$  an infinite ordinal are of particular significance to our work; the paper [GSW26] was an investigation into the partition relations which hold on these orders. They are ordered by the *lexicographic order*  $<_{\text{lex}}$ , where for  $x, y \in {}^\alpha 2$ ,

$$x <_{\text{lex}} y \iff x(\delta) < y(\delta),$$

where  $\delta = \delta(x, y)$  is the minimal ordinal  $< \alpha$  on which the sequences  $x$  and  $y$  disagree.

The set  ${}^{<\omega} 2$  of finite binary sequences is also very important to our work; we equip it with a tree ordering and a linear ordering. The tree  $\langle {}^{<\omega} 2, \sqsubseteq \rangle$  is ordered by  $\sqsubseteq$ , where  $x \sqsubseteq y$  iff  $x$  is an initial segment of  $y$ , i.e.  $y \upharpoonright |x| = x$ . The linear order  $\langle {}^{<\omega} 2, <_{\text{lex}} \rangle$  is obtained by extending the definition of the lexicographic ordering so that  $x <_{\text{lex}} y$  includes the cases  $x \frown \langle 1 \rangle \sqsubseteq y$  and  $x \sqsupseteq y \frown \langle 0 \rangle$ ; this makes  $\langle {}^{<\omega} 2, <_{\text{lex}} \rangle$  a linear order ordered as  $\eta$ .

## 2 Consistency results

We begin with the key motivating result which led to the work done in this paper:

**Fact 2.** *Let  $\tau$  be a countable linear order type such that either  $\omega\omega^* \leq \tau$  or  $\omega^*\omega \leq \tau$ . Then for any ordinal  $\alpha$ ,*

$$\langle {}^\alpha 2, <_{\text{lex}} \rangle \not\prec (\tau)^\tau.$$

This is a consequence of Theorems 1 and 2 of [GSW26], corresponding to the case in which  $\tau$  is non-scattered and the case in which  $\tau$  is scattered, respectively:

**Theorem.** (Theorem 1, [GSW26]) *Let  $\tau \neq 0$  be an order type with  $\tau + \tau \leq \tau$ . Then for any ordinal  $\alpha$ ,*

$$\langle {}^\alpha 2, <_{\text{lex}} \rangle \not\prec (\tau)^\tau.$$

**Theorem.** (Theorem 2, [GSW26]) *Let  $\alpha$  be an ordinal and  $\tau$  a scattered, well-orderable order type. If  $\omega\omega^* \leq \tau$  or  $\omega^*\omega \leq \tau$  then*

$$\langle {}^\alpha 2, <_{\text{lex}} \rangle \not\prec (\tau)^\tau.$$

We show now that Fact 2 does not generalise to all linear orders:

**Theorem 3.** *It is consistent with ZF that there exists a linear order  $\langle L, < \rangle$  such that for any countable  $\tau$ ,*

$$\langle L, < \rangle \rightarrow (\tau)^\tau.$$

We first prove the consistency of this statement with ZFA by means of permutation models, and then show that the result lifts to a ZF result using symmetric systems.<sup>3</sup>

**Lemma 4.** *It is consistent with ZFA that there exists a linear order  $\langle L, < \rangle$  such that for any countable  $\tau$ ,*

$$\langle L, < \rangle \rightarrow (\tau)^\tau.$$

*Proof.* Let  $M$  be a model of ZFCA + CH whose set of atoms  $A$  has cardinality  $\aleph_1$ , and let  $\prec \subseteq A \times A$  be such that  $\langle A, \prec \rangle$  is a saturated linear order (this is possible as  $M \models 2^{\aleph_0} = \aleph_1$ : see [Ma06, Corollary 4.3.13]). Let  $\mathcal{G} \leq \text{Aut}(A)$  be the group of all permutations of  $A$  which preserve the order  $\prec$ , i.e. the group of all order-automorphisms of  $\langle A, \prec \rangle$ , and let  $\mathcal{F} \subseteq \mathcal{P}(\mathcal{G})$  be the normal filter generated by subgroups of  $\mathcal{G}$  which fix countable subsets of  $A$ . Write  $N \subseteq M$  for the permutation model generated by  $\mathcal{G}$  and  $\mathcal{F}$ . It is clear that  $\langle A, \prec \rangle \in N$ ; observe that all countable subsets of  $A$  in  $M$  are also elements of  $N$ , as for  $E \subseteq A$  countable,  $\text{fix}(E) \leq \text{sym}(E)$ , so  $\text{sym}(E) \in \mathcal{F}$ . We claim that in  $N$ , the order  $\langle A, \prec \rangle$  has the desired property.

Let  $\tau$  be a countable order type and let  $F : [\langle A, \prec \rangle]^\tau \rightarrow 2$  be a colouring in  $N$ . Since  $F$  is symmetric, it has some support  $E \subseteq A$ , a countable set of atoms. Then by the  $\omega_1$ -saturation of  $\langle A, \prec \rangle$  in  $M$ , we can find (in  $M$ ) an interval  $I$  of  $\langle A, \prec \rangle$  such that  $I \cap E = \emptyset$ . Then let  $H \in [I]^\tau$  be arbitrary; by the above,  $H \in N$ , and  $M$  and  $N$  agree on  $[H]^\tau$ . We claim that  $H$  is homogeneous for  $F$ . Let  $H' \in [H]^\tau$ , and let  $i : E \cup H \rightarrow E \cup H'$  be a partial automorphism of  $A$  which fixes every element of  $E$  and which maps  $H$  to  $H'$  order-preservingly, noting that such an  $i$  exists as the convex hull of  $H$  is disjoint from  $E$ . Since  $\langle A, \prec \rangle$  is saturated,  $i$  can be extended to a full order-automorphism,  $\pi$ , say; treating  $\pi$  as an automorphism of  $N$ , we have  $\pi(F)(\pi(H)) = F(H)$ , but  $\pi(F) = F$  as  $\pi \in \text{fix}(E)$  and  $\pi(H) = H'$  by construction, so  $F(H') = F(H)$ . Since  $H' \in [H]^\tau$  was arbitrary, it follows that  $H$  is homogeneous for  $F$ .  $\square$

*Proof of Theorem 3.* For  $\langle L, < \rangle$  a linear order,  $\tau$  an order type, the statement

$$\langle L, < \rangle \rightarrow (\tau)^\tau$$

is determined by  $\mathcal{P}^3(L)$ , as  $< \in \mathcal{P}^3(L)$ , any  $H \subseteq L$  is of course an element of  $\mathcal{P}(L)$ , and any colouring  $F : [\langle L, < \rangle]^\tau \rightarrow 2$  can be encoded as a set of subsets of  $L$ , i.e. an element of  $\mathcal{P}^2(L)$ .

Now, let  $M$ ,  $\langle A, \prec \rangle$ , and  $N$  be as in the proof of Lemma 4, and let  $V \subseteq M$  be the kernel of  $M$ , i.e. the class of hereditary sets, so  $V \models \text{ZFC}$ . Then by the

<sup>3</sup>ZF(C)A is the theory of *Zermelo-Fraenkel set theory with atoms*; see [Je73, §4] for a detailed treatment of this theory and of the theory of permutation models.

Jech-Sochor theorem (cf. [Je73, Theorem 6.1]), there is a symmetric extension  $V(G) \models \text{ZF}$  of  $V$  and a set  $\bar{A} \in V(G)$  such that

$$(\mathcal{P}^3(A))^N \text{ is } \in\text{-isomorphic to } (\mathcal{P}^3(\bar{A}))^{V(G)}.$$

It follows that in  $V(G)$ , there is an ordering  $\prec' \subseteq \bar{A} \times \bar{A}$  such that for all countable  $\tau$

$$\langle \bar{A}, \prec' \rangle \rightarrow (\tau)^\tau. \quad \square$$

The Jech-Sochor theorem uses sets of sets of ordinals in a generic extension to mimic the role of atoms in ZFA, so the order  $\langle \bar{A}, \prec' \rangle$  built above consists of sets of sets of ordinals. The question of whether this is necessary or whether one can consistently find an  $\langle L, < \rangle$  with the same property such that  $L$  is a set of ordinals inspired the work of the next section.

We remark here that an almost identical proof to the above yields the following result also:<sup>4</sup>

**Proposition 5.** *It is consistent with ZF that there exists a graph  $G$  such that for any countable graph  $\Gamma$ ,*

$$G \rightarrow (\Gamma)^\Gamma.$$

### 3 The failure of weak choice principles

The particular fragments of choice with which we concern ourselves in this paper are  $\text{KWP}_1$  and  $\text{O}$ , which we introduce below:

**Definition 6.** *The Kinna-Wagner Selection Principle  $\text{KWP}_1$  is the statement that every set injects into the power set of an ordinal,*

$$\forall A \exists \alpha \in \text{Ord} \exists f : A \rightarrow \mathcal{P}(\alpha) \text{ injective.}$$

Some comments: this principle was introduced in [KW55], originally as a different statement which was shown equivalent to the formulation given above. More generally one can study the principle  $\text{KWP}_\beta$  for  $\beta$  an ordinal, which is the statement that  $\forall A \exists \alpha \in \text{Ord} \exists f : A \rightarrow \mathcal{P}^\beta(\alpha)$  injective, so the Axiom of Choice is  $\text{KWP}_0$ . The statement  $\text{KWP}$  then denotes the statement that  $\text{KWP}_\beta$  holds for some  $\beta$ ; see [KS26]. In this paper we concern ourselves only with  $\text{KWP}_1$ .

**Definition 7.** *The Ordering Principle  $\text{O}$  is the statement that every set can be linearly ordered, i.e. for any  $A$  there is a relation  $< \subseteq A \times A$  which is transitive, irreflexive, and trichotomous on  $A$ .*

**Observation 8.**

$$\text{KWP}_1 \implies \text{O}.$$

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<sup>4</sup>We modify the proof of Lemma 4 in the following way. Start with a model of  $\text{ZFCA} + \text{CH}$  whose set of atoms  $A$  can be equipped with an edge relation making it a saturated graph; where in that proof we found an interval  $I$  disjoint from the countable support set  $E$ , in this proof we appeal to  $\omega_1$ -saturation to find a copy of  $R$  disjoint from  $E$ .

*Proof.* Assume  $\text{KWP}_1$ , and let  $A$  be a set. Then there is an ordinal  $\alpha$  and an injection  $i : A \hookrightarrow {}^\alpha 2$ . It follows that the relation  $<$  defined on  $A$  by, for  $a, b \in A$ ,

$$a < b \iff i(a) <_{\text{lex}} i(b)$$

is a linear order on  $A$ . □

The reverse implication does not hold; see [Fe71] or [Pi72].

We now return to the structural IEPs shown consistent in §2 and show that the existence of a structure satisfying such a partition relation yields a failure of one of these choice principles. For the purposes of the proof it will be easier to phrase our results contrapositively, i.e. in the form “if the structure  $A$  does not witness the failure of some choice principle, then  $A \not\vdash (C)^C$ ”. In order to build our colourings with no homogeneous sets, we make use of the following concept from [GSW26]:

**Definition 9.** Let  $A, C$  be structures with  $[A]^C \neq \emptyset$ . A set  $\mathcal{D} \subseteq [A]^C$  is dense if for any  $X \in [A]^C$ , there is some  $X' \in [X]^C$  with  $X' \in \mathcal{D}$ .

A colouring of some dense  $\mathcal{D}$  with no homogeneous set extends to a colouring of the entirety of  $[A]^C$  with no homogeneous set, e.g. by just sending every element of  $[A]^C \setminus \mathcal{D}$  to 0.

**Observation 10.** Let  $A, C$ , be structures, let  $\mathcal{D} \subseteq [A]^C \neq \emptyset$  be dense, and let  $F : \mathcal{D} \rightarrow 2$  have no homogeneous set, in the sense that for any  $X \in \mathcal{D}$ , there is some  $X' \in [X]^C \cap \mathcal{D}$  with  $F(X) \neq F(X')$ . Then  $F$  extends to a colouring  $F' : [A]^C \rightarrow 2$  witnessing  $A \not\vdash (C)^C$ . □

**Theorem 11.** Let  $\tau$  be a countable non-scattered order type, and let  $\langle L, < \rangle$  be a linear order. If the set  $L$  injects into the power set of an ordinal, then

$$\langle L, < \rangle \not\vdash (\tau)^\tau.$$

Equivalently, if the linear order  $\langle L, < \rangle$  witnesses  $\langle L, < \rangle \rightarrow (\tau)^\tau$  for  $\tau$  some countable non-scattered order type, then the set  $L$  witnesses the failure of  $\text{KWP}_1$ . We first prove a lemma of independent interest:

**Lemma 12.** Let  $\prec$  be an arbitrary linear order on  $\mathbb{Q}$ , and let  $<$  denote the usual linear order on  $\mathbb{Q}$ . Then there exists some  $A \in [(\mathbb{Q}, <)]^\eta$  satisfying one of the following conditions:

1.  $\langle A, \prec \rangle \cong \omega$ ;
2.  $\langle A, \prec \rangle \cong \omega^*$ ;
3.  $\prec$  is identical to  $<$  on  $A$ ;
4.  $\prec$  is the reverse of  $<$  on  $A$ .

*Proof of Lemma 12.* Fix such an ordering  $\prec$ . For the duration of this proof we fix  $\langle q_n : n \in \omega \rangle$  an enumeration of  $\mathbb{Q}$ ; when we refer to a rational “of minimal index”, we mean  $q_n$  with  $n$  minimal according to this enumeration.

First observe that whenever  $\langle L, < \rangle$  is ordered as  $\eta$  and  $A \subseteq L$  is scattered under  $<$ ,  $\langle L \setminus A, < \rangle$  is still ordered as  $\eta$ ; otherwise, it would have to be the case that  $A$  contained an interval of  $\langle L, < \rangle$  (either between two elements of  $L$  or an initial or final segment); but then  $A$  is not scattered, a contradiction.

**Claim 12.1.** If  $\prec$  is such that whenever  $\langle I, \prec \rangle$  is a proper initial segment of  $\langle \mathbb{Q}, \prec \rangle$ ,  $\langle I, < \rangle$  is scattered, then there is  $A \in [\langle \mathbb{Q}, < \rangle]^\eta$  with  $\langle A, \prec \rangle \cong \omega$ .

*Proof of claim.* Note that the given condition implies that  $\langle \mathbb{Q}, \prec \rangle$  has no maximal element and thus has cofinality  $\omega$ . Fix a cofinal sequence  $x_0 \prec x_1 \prec \dots$  in  $\langle \mathbb{Q}, \prec \rangle$ , and for each  $n \in \omega$  set  $F_n := \{y \in \mathbb{Q} : x_n \preceq y\}$ . Then each  $F_n$  is dense in  $\langle \mathbb{Q}, < \rangle$  (as  $\mathbb{Q} \setminus F_n$  is scattered under  $<$ ). We can now run a back-and-forth construction to build  $A \in [\langle \mathbb{Q}, < \rangle]^\eta$  with the desired property.

We build  $\langle a_n : n \in \omega \rangle$  recursively like so: take  $a_0 \in F_0$ , and, given  $a_0, \dots, a_{n-1}$ , take  $a_n \in F_n$  such that for all  $m < n$ ,

$$\begin{aligned} a_n < a_m &\iff q_n < q_m, \text{ and} \\ a_n > a_m &\iff q_n > q_m; \end{aligned}$$

then set  $A := \{a_n : n \in \omega\}$ .<sup>5</sup> Clearly  $\langle A, < \rangle \cong \langle \mathbb{Q}, < \rangle$ , so  $A \in [\langle \mathbb{Q}, < \rangle]^\eta$ . Now, since for each  $n \in \omega$ ,  $A$  contains only finitely many elements of  $F_n \setminus F_{n+1}$ ,  $\langle A, \prec \rangle$  is ordered as  $\omega$ . ■ Claim

By symmetry, the corresponding statement with initial segments replaced by final segments and the order type  $\omega$  replaced by  $\omega^*$  also holds.

**Claim 12.2.** Suppose  $A \in [\langle \mathbb{Q}, < \rangle]^\eta$  has the property that for all  $A' \in [\langle A, < \rangle]^\eta$ ,  $\langle A', \prec \rangle \not\cong \omega$  and  $\langle A', \prec \rangle \not\cong \omega^*$ . Then there exists  $B \in [\langle A, < \rangle]^\eta$  with the following property:

For any  $I \subseteq B$  such that  $\langle I, \prec \rangle$  is either an initial or final segment of  $\langle B, \prec \rangle$ ,  $\langle I, < \rangle$  is non-scattered.

Moreover, this  $B$  is determined from  $A$  in a canonical way.

*Proof of claim.* Suppose some (necessarily proper) initial segment  $I$  of  $\langle A, \prec \rangle$  has that  $\langle I, < \rangle$  is scattered. Taking a union of all such  $\prec$ -initial segments, we obtain  $J \subseteq A$ , the maximal initial segment of  $\langle A, \prec \rangle$  with the property that for any proper initial segment  $I$  of  $\langle J, \prec \rangle$ ,  $\langle I, < \rangle$  is scattered.

If  $\langle J, < \rangle$  is non-scattered, i.e.  $\eta \leq \text{otp}\langle J, < \rangle$ , then letting  $T \in [\langle J, < \rangle]^\eta$  and applying Claim 12.1 with  $\langle T, < \rangle$  in place of  $\langle \mathbb{Q}, < \rangle$ , we obtain  $A' \in [\langle T, < \rangle]^\eta$  with  $\langle A', \prec \rangle \cong \omega$ , contradicting our assumption that no  $A' \in [\langle A, < \rangle]^\eta$  has this property.

So  $\langle J, < \rangle$  is scattered. It follows that  $B' := A \setminus J$  is ordered as  $\eta$  and has the property that for any  $I$  an initial segment of  $\langle B', \prec \rangle$ ,  $\langle I, < \rangle$  is non-scattered.

<sup>5</sup>Observe that this does not use Choice, as we can at each step take  $a_n$  to be minimal according to the enumeration  $\langle q_n : n \in \omega \rangle$ .

Running the symmetric procedure on  $B'$ , we reduce to  $B \in [\langle B', < \rangle]^n \subseteq [\langle A, < \rangle]^n$  with the property that whenever  $I$  is an initial segment *or* a final segment of  $\langle B, \prec \rangle$ ,  $\langle I, < \rangle$  is non-scattered.

Note finally that  $B$  was obtained from  $A$  by removing uniquely-determined initial and final segments of  $\langle A, \prec \rangle$ , and so this procedure canonically determines  $B$ . ■ Claim

If there exists some  $A \in [\langle \mathbb{Q}, < \rangle]^n$  with  $\langle A, \prec \rangle \cong \omega$  or  $\langle A, \prec \rangle \cong \omega^*$  we are done, so assume that no such  $A$  exists. We now define two functions

$$\begin{aligned} q : {}^{<\omega}2 &\rightarrow \mathbb{Q}, \\ Q : {}^{<\omega}2 &\rightarrow [\langle \mathbb{Q}, < \rangle]^n \end{aligned}$$

recursively; our desired  $A$  will be a particular subset of  $q'' {}^{<\omega}2$ .

Set  $q(\emptyset)$  to be any element of  $\mathbb{Q}$  (e.g.  $q_0$  in the enumeration of  $\mathbb{Q}$  fixed above), and  $Q(\emptyset) = \mathbb{Q}$ . For  $s \in {}^{<\omega}2$ , given  $q(s)$  and  $Q(s)$ , we find  $Q(s \smallfrown \langle 0 \rangle)$  and  $Q(s \smallfrown \langle 1 \rangle)$  in the following way:

First, apply Claim 12.2 to  $Q(s) \cap (-\infty, q(s))_{<}$  and  $Q(s) \cap (q(s), \infty)_{<}$  to obtain  $B_0$  and  $B_1$ , respectively, each with the property that any  $\prec$ -initial segment or  $\prec$ -final segment is non-scattered in the order  $<$ . Let  $n \in \omega$  be minimal such that either

- (a)  $B'_0 := B_0 \cap (\leftarrow, q_n)_{<}$  and  $B'_1 := B_1 \cap (q_n, \rightarrow)_{<}$  are both non-empty, or
- (b)  $B''_0 := B_0 \cap (q_n, \rightarrow)_{<}$  and  $B''_1 := B_1 \cap (\leftarrow, q_n)_{<}$  are both non-empty.

In case (a),  $\langle B'_0, < \rangle$  and  $\langle B'_1, < \rangle$  are both non-scattered, and as such each consist of a densely-ordered sum of scattered intervals; by taking the minimal-indexed element of each of these scattered intervals, then removing the minimal or maximal element if necessary, we can find  $A_0 \in [\langle B'_0, < \rangle]^n$ ,  $A_1 \in [\langle B'_1, < \rangle]^n$  in a canonical way; by construction,  $A_0 \prec A_1$ . In case (b), we instead obtain  $A_0 \in [\langle B''_0, < \rangle]^n$ ,  $A_1 \in [\langle B''_1, < \rangle]^n$  with  $A_0 \succ A_1$ .<sup>6</sup> Set  $Q(s \smallfrown \langle 0 \rangle) = A_0$  and  $Q(s \smallfrown \langle 1 \rangle) = A_1$ , and let  $q(s \smallfrown \langle 0 \rangle) \in Q(s \smallfrown \langle 0 \rangle)$ ,  $q(s \smallfrown \langle 1 \rangle) \in Q(s \smallfrown \langle 1 \rangle)$  of minimal index.

We now find our desired  $A$  as a subset of  $q'' {}^{<\omega}2$ . Consider the colouring  $c : {}^{<\omega}2 \rightarrow 2$  given by, for  $s \in {}^{<\omega}2$ ,

$$c(s) = \begin{cases} 0 & \text{if } Q(s \smallfrown \langle 0 \rangle) \prec Q(s \smallfrown \langle 1 \rangle); \\ 1 & \text{if } Q(s \smallfrown \langle 0 \rangle) \succ Q(s \smallfrown \langle 1 \rangle). \end{cases}$$

By, e.g. the Halpern–Läuchli theorem in the setting of one tree, there exists  $H \subseteq {}^{<\omega}2$  such that  $\langle H, \sqsubseteq \rangle$  is isomorphic to  $\langle {}^{<\omega}2, \sqsubseteq \rangle$  on which  $c$  is homogeneous; then  $A := q'' H$  has the desired property: if  $H$  is homogeneous with colour 0, then for any  $s, t \in {}^{<\omega}2$  with  $s <_{\text{lex}} t$ , we have that  $q(s) \prec q(t)$ , so  $A$  satisfies condition 3; if instead  $H$  is homogeneous with colour 1,  $s <_{\text{lex}} t$  implies  $q(s) \succ q(t)$ , and so  $A$  satisfies condition 4. □

<sup>6</sup>It is possible that both (a) and (b) hold; in this case we proceed as in case (a).

*Proof of Theorem 11.* Fix  $\tau$ ,  $\langle L, < \rangle$ , and an injection  $i : L \hookrightarrow {}^\alpha 2$  for some ordinal  $\alpha$ . We will build a colouring with no homogeneous set on a dense subset of  $[\langle L, < \rangle]^\tau$ .

For  $j \in \{0, 1\}$ , define  $\mathcal{D}_j \subseteq [\langle L, < \rangle]^\tau$  like so:

$$\begin{aligned}\mathcal{D}_0 &:= \{A \in [\langle L, < \rangle]^\tau : i \restriction A \text{ is ordered as } \omega \text{ or } \omega^* \text{ under } <_{\text{lex}}\}; \\ \mathcal{D}_1 &:= \{A \in [\langle L, < \rangle]^\tau : i \text{ is order-preserving or order-reversing on } A\},\end{aligned}$$

where here “order-preserving” (resp. “reversing”) means with respect to the orders  $\langle L, < \rangle$  and  $\langle {}^\alpha 2, <_{\text{lex}} \rangle$ , and define  $\mathcal{D} := \mathcal{D}_0 \cup \mathcal{D}_1$ . Observe that each of  $\mathcal{D}_0, \mathcal{D}_1$  are downwards closed.

**Claim 11.1.**  $\mathcal{D}$  is dense in  $[\langle L, < \rangle]^\tau$ .

*Proof of claim.* Let  $A \in [\langle L, < \rangle]^\tau$ ; we will show that there is some  $A' \in [A]^\tau$  with  $A' \in \mathcal{D}$ . Note that  $A$  can be reduced to some  $Q \in [A]^\eta$ ; then applying Lemma 12 to  $Q$  with the orders  $<$  and  $\prec := i^{-1}(<_{\text{lex}})$ , we reduce further to some  $Q' \in [Q, <]^\eta$  such that either  $i \restriction Q'$  is ordered as one of  $\omega$  or  $\omega^*$ , or  $i$  is order-preserving or order-reversing on  $Q'$ . Now let  $A'$  be any element of  $[\langle Q', < \rangle]^\tau$  (these exist as  $\langle Q', < \rangle$  is ordered as  $\eta$ ); it is clear that  $A'$  retains whichever of the above properties  $Q'$  had under  $i$ . ■ Claim

Now define  $c : \mathcal{D} \rightarrow 2$  like so: for  $A \in \mathcal{D}_0$ , write  $\{a_n : n \in \omega\}$  for the natural enumeration of the elements of  $A$  determined by the ordering of  $i \restriction A$  in type  $\omega$  or  $\omega^*$ ; then set

$$c(A) = \begin{cases} 0 & \text{if } a_0 > a_1; \\ 1 & \text{if } a_0 < a_1. \end{cases}$$

To see that no  $A \in \mathcal{D}_0$  can be homogeneous for this, note that since  $\tau$  is non-scattered, we can find  $n < m$  such that  $a_n < a_m$  and each of  $I_0 := (\leftarrow, a_n)_{<} \cap A$ ,  $I_1 := (a_n, a_m)_{<} \cap A$  and  $I_2 := (a_m, \rightarrow)_{<} \cap A$  are non-scattered. Letting  $B := \{a_k : k < m, k \neq n\}$ , which is finite, for each  $l < 3$ , we also have that  $I_l \setminus B$  is non-scattered, which further contains a copy  $J_l$  of  $I_l$ . It follows that  $A' := J_0 \cup J_1 \cup J_2 \cup \{a_n, a_m\}$  is isomorphic to  $A$  and  $c(A') = 0$ . Similarly, we can achieve that  $c(A') = 1$ .

For  $A \in \mathcal{D}_1$ , we simply apply the colouring from [GSW26, Theorem 1] to  $i \restriction A$ ; as this colouring witnesses that  $\langle {}^\alpha 2, <_{\text{lex}} \rangle \not\prec (\tau)^\tau$  (or that  $\langle {}^\alpha 2, <_{\text{lex}} \rangle \not\prec (\tau^*)^{\tau^*}$ ) it is clear that there is no  $A \in \mathcal{D}_1$  homogeneous for this. □

**Theorem 13.** *Let  $\tau$  be a well-orderable scattered order type with  $\omega\omega^* \leq \tau$  or  $\omega^*\omega \leq \tau$ , and let  $\langle L, < \rangle$  be a linear order. If the set  $L$  injects into the power set of an ordinal, then*

$$\langle L, < \rangle \not\prec (\tau)^\tau.$$

We will first prove the special case of this theorem where  $\tau$  is additively indecomposable, then complete the proof of Theorem 13 by explaining how to adapt this proof to the general case.

**Lemma 14.** *Let  $\tau$  be a well-orderable additively indecomposable scattered order type with either  $\omega\omega^* \leq \tau$  or  $\omega^*\omega \leq \tau$ , and let  $\langle L, < \rangle$  be a linear order. If the set  $L$  injects into the power set of an ordinal, then*

$$\langle L, < \rangle \not\rightarrow (\tau)^\tau.$$

We first introduce some definitions and notation used throughout the proofs of Lemma 14 and Theorem 13. Particular sets ordered as  $\omega$  or  $\omega^*$  will be of key importance throughout; we adopt the convention that if  $X \in [\langle L, < \rangle]^\omega$  for some  $\langle L, < \rangle$ ,  $\langle x_n : n \in \omega \rangle$  is the natural increasing enumeration of  $X$ , so  $X = \{x_n : n \in \omega\}$  and  $x_0 < x_1 < \dots$ . Similarly, if  $X \in [\langle L, < \rangle]^{\omega^*}$ ,  $\langle x_n : n \in \omega \rangle$  is the natural decreasing enumeration of  $X$ . We now introduce a useful notion from [GSW26]:

For  $X \in [\langle {}^\alpha 2, <_{\text{lex}} \rangle]^\omega$ , say that  $X$  is *canonised* if for  $x, y \in X$  with  $x < y$ , the value of  $\delta(x, y)$  is determined by  $x$ ; symmetrically, for  $X \in [\langle {}^\alpha 2, <_{\text{lex}} \rangle]^{\omega^*}$ , say that  $X$  is canonised if the value of  $\delta(x, y)$  for  $x < y \in X$  is determined by  $y$ . In either case, this is equivalent to the condition that  $\delta(x_0, x_1) < \delta(x_1, x_2) < \delta(x_2, x_3) < \dots$ .

**Observation 15.** *Given any  $X \in [\langle {}^\alpha 2, <_{\text{lex}} \rangle]^\omega \cup [\langle {}^\alpha 2, <_{\text{lex}} \rangle]^{\omega^*}$ , there is  $X' \in [X]^{\text{otp}(X)}$  which is canonised.*<sup>7</sup>  $\square$

The proofs start by assuming the existence of some injection  $i : L \hookrightarrow {}^\alpha 2$ . Given a fixed such  $i$ , for  $O \in [\langle L, < \rangle]^\omega$  or  $O \in [\langle L, < \rangle]^{\omega^*}$ , say  $O$  is *i-canonised* if the following two conditions hold:

1.  $i$  is order-preserving or order-reversing on  $O$ ;
2.  $i \restriction O$  is canonised in  $\langle {}^\alpha 2, <_{\text{lex}} \rangle$ .

Throughout both proofs the splitting levels (in  $\langle {}^\alpha 2, <_{\text{lex}} \rangle$ ) between elements of  $i \restriction A$  for  $A \in [\langle L, < \rangle]^\tau$  will be significant, so we introduce the following notation to avoid clutter: for  $x, y \in L$ , write

$$\delta_i(x, y) := \delta(i(x), i(y)).$$

In particular, it will often be important for us to consider two such consecutive quantities in copies of  $\omega$  or  $\omega^*$ : for  $X \in [\langle L, < \rangle]^\omega \cup [\langle L, < \rangle]^{\omega^*}$  and  $n \in \omega$ , write

$$d_n(X) := (\delta_i(x_n, x_{n+1}), \delta_i(x_{n+1}, x_{n+2})).$$

For  $n \in \omega$  fixed, the  $d_n(X)$  are ordered according to the lexicographic ordering on  $\alpha \times \alpha$ . The quantity  $d_2(X)$  will be important in one case in the proof of Lemma 14, and the quantity  $d_0(X)$  will be important in the proof of Theorem 13.

*Proof of Lemma 14.* Fix  $\tau$ ,  $\langle L, < \rangle$ , and an injection  $i : L \hookrightarrow {}^\alpha 2$  for some ordinal  $\alpha$ . We appeal here to a result of Jullien:

<sup>7</sup>In fact, one can define a uniform procedure for picking out such an  $X'$  canonically: see [GSW26, Lemma 9].

**Fact 15.1.** ([Ju68]) Any indecomposable scattered order is either strictly indecomposable to the right or to the left.<sup>8</sup>

Wlog  $\tau$  is strictly indecomposable to the right. We will define a colouring  $c : \mathcal{D} \rightarrow 2$  on a dense subset  $\mathcal{D}$  of  $[(L, <)]^\tau$  with no homogeneous set by means of the following strategy:  $\mathcal{D}$  will have the property that every  $A \in \mathcal{D}$  has an interval which can be picked out and which is followed by (at least) two consecutive elements of  $A$ . Call this the *distinguished interval* of  $A$ ; roughly speaking, we first colour according to whether or not  $i$  is order-preserving on the two elements of  $A$  immediately following the distinguished interval. Since  $\tau$  is strictly indecomposable to the right, for any  $x < y \in A$  both above the distinguished interval, we can reduce to some  $A' \in [A]^\tau$  such that  $x$  and  $y$  are the two elements of  $A'$  picked out in this way; it follows that for  $A$  to be homogeneous for this colouring, it must be the case that  $i$  is either order-preserving or order-reversing on the final segment of  $A$  above the distinguished interval. We then colour those  $A$  with this property by applying the colouring  $C$  from [GSW26, Theorem 2] to the image under  $i$  of this final segment of  $A$ ; since  $i$  is either order-preserving or order-reversing on this final segment, its image in  $\langle \alpha 2, <_{\text{lex}} \rangle$  is ordered as some order type  $\varphi$  with the property that  $C$  witnesses  $\langle \alpha 2, <_{\text{lex}} \rangle \not\prec (\varphi)^\varphi$ . It therefore follows that none of these sets will be homogeneous for  $c$  either.

**Case 1.**  $\tau$  has a condensation class ordered as  $\omega^*$  or a condensation class ordered as  $\zeta$ .

Here our distinguished interval (and the dense subset of  $[(L, <)]^\tau$  on which we can guarantee that such an interval exists) has a somewhat complex definition. Let us write

$$[A]_c^{\omega^*} := \{X \in [A]^{\omega^*} : X \text{ is convex in } A\}.$$

Say that  $X \in [A]_c^{\omega^*}$  is *almost  $i$ -canonised* if  $X \setminus \{x_0, x_1\}$  is  $i$ -canonised, and associate each such  $X$  with the quantity  $d_2(X) = (\delta_i(x_2, x_3), \delta_i(x_3, x_4))$ .

**Claim 15.2.** For any  $A \in [(L, <)]^\tau$ , there is  $A' \in [A]^\tau$  with the following two properties:

1. There is at least one  $X \in [A']_c^{\omega^*}$  which is almost  $i$ -canonised;
2. Amongst all almost  $i$ -canonised  $X \in [A]_c^{\omega^*}$  on which  $d_2(X)$  is minimal, there is a *leftmost* one (i.e. one which is  $<$ -minimal).

*Proof of claim.* First observe that for any  $A \in [(L, <)]^\tau$ , there is  $A^* \in [A]^\tau$  with at least one  $X \in [A^*]_c^{\omega^*}$  which is almost  $i$ -canonised: let  $Y = \{y_n : n \in \omega\} \in [A]_c^{\omega^*}$  be arbitrary. Then  $Y \setminus \{y_0, y_1\}$  is also ordered as  $\omega^*$  in  $\langle L, < \rangle$ , and as such there is some  $Y' \in [Y]^{\omega^*}$  on which  $i$  is order-preserving or order-reversing, e.g.

<sup>8</sup>This result was proved in ZFC but still holds in ZF for well-orderable scattered orders; see [GSW26, Footnote 6].

by Ramsey's theorem. Reduce further to some  $Y'' \in [Y']^{\omega^*}$  such that  $i \restriction Y''$  is canonised. Set

$$A^* := (A \setminus Y) \cup Y'' \cup \{y_0, y_1\}.$$

Then  $Y'' \cup \{y_0, y_1\} \in [A^*]_c^{\omega^*}$  is almost  $i$ -canonised. Write

$$\mathcal{A} := \{B \in [A]^\tau : \text{at least one } X \in [B]_c^{\omega^*} \text{ is almost } i\text{-canonised}\};$$

it follows from the above that  $\mathcal{A}$  is non-empty. For  $B \in \mathcal{A}$ , write

$$d_2^{\min}(B) := \min\{d_2(X) : X \in [B]_c^{\omega^*} \text{ almost } i\text{-canonised}\},$$

and let  $A^\dagger$  be an element of  $\mathcal{A}$  which minimises  $d_2^{\min}(A^\dagger)$ . Then there is at least one almost  $i$ -canonised  $X \in [A^\dagger]_c^{\omega^*}$  such that  $d_2(X) = d_2^{\min}(A^\dagger)$  is minimal, but this is not necessarily unique. We further reduce to  $A' \in [A^\dagger]^\tau$  with  $d_2^{\min}(A') = d_2^{\min}(A^\dagger)$ , with a leftmost witness  $X$ , in the following way: pick some almost  $i$ -canonised  $X \in [A^\dagger]_c^{\omega^*}$  with  $d_2(X) = d_2^{\min}(A^\dagger)$ , and reduce to  $A' \in [A^\dagger]^\tau$  by, for all almost  $i$ -canonised  $Z \in [A^\dagger]_c^{\omega^*}$  with  $d_2(Z) = d_2^{\min}(A^\dagger)$  and  $Z < X$ , removing  $z_3$ . Observe that this is well-defined, as for any  $Z_0, Z_1 \in [A^\dagger]_c^{\omega^*}$ , if  $d_2(Z_0) = d_2(Z_1)$  then either  $Z_0 = Z_1$  or  $Z_0 \cap Z_1 = \emptyset$ , and  $A'$  still has order type  $\tau$  as we have simply replaced some of its intervals ordered as  $\omega^*$  by proper subsets also ordered as  $\omega^*$ .

Now  $X \in [A']_c^{\omega^*}$  still has  $d_2(X) = d_2^{\min}(A') = d_2^{\min}(A^\dagger)$ , which is minimal, but for any other almost  $i$ -canonised  $Z \in [A']_c^{\omega^*}$  with  $d_2(Z) = d_2(X)$  and  $Z < X$ ,  $Z$  has been replaced by  $Z' := Z \setminus \{z_3\}$ , and  $d_2(Z') = (\delta_i(z_2, z_4), \delta_i(z_4, z_5))$ . This  $d_2(Z')$  is strictly above  $d_2(X)$  in the lexicographic ordering of  $\alpha \times \alpha$ ; this is because  $\delta_i(z_2, z_4) = \min\{\delta_i(z_2, z_3), \delta_i(z_3, z_4)\} = \delta_i(z_2, z_3)$ , and  $\delta_i(z_4, z_5)$  must be strictly greater than  $\delta_i(z_3, z_4)$ , by the  $i$ -canonisation of  $Z \setminus \{z_0, z_1\}$ . It follows that  $A'$  is as described. ■ Claim

Write  $\mathcal{E} \subseteq [L, <]^\tau$  for the set of those  $A \in [L, <]^\tau$  satisfying the property described in Claim 15.2; our dense set  $\mathcal{D} \subseteq [L, <]^\tau$  in this case is the set of those  $A \in \mathcal{E}$  satisfying the further condition that  $d_2^{\min}(A)$  is minimal in  $\{d_2^{\min}(A') : A' \in [A]^\tau \cap \mathcal{E}\}$ , and the distinguished interval of such an  $A$  is the leftmost almost  $i$ -canonised  $X \in [A]_c^{\omega^*}$  with  $d_2(X)$  minimal.  $\dashv$  Case 1

**Case 2.** All of the infinite condensation classes in  $\tau$  are ordered as  $\omega$ .

In this case, the definitions of both the dense set and the distinguished interval are less complex. First note the following:

**Claim 15.3.**  $\tau$  contains two consecutive condensation classes ordered as  $\omega$ .

*Proof of claim.* First note that since  $\tau$  is scattered, every infinite interval of  $\tau$  contains an infinite condensation class; an infinite order whose condensation classes are all finite is non-scattered, as e.g. the suborder consisting of the leftmost element of each condensation class is densely ordered. Since  $\omega\omega^* \leq \tau$  or  $\omega^*\omega \leq \tau$ , it can certainly be split into multiple infinite intervals, each of which will contain at least one infinite condensation class.

Fix some  $A$  ordered as  $\tau$  and let  $X_0 < X_1$  be distinct condensation classes of  $A$  ordered as  $\omega$ . If  $X_0$  and  $X_1$  are not consecutive, there is a non-empty interval of  $A$  between them, which must be infinite, as otherwise it would be part of  $X_1$ ; we therefore have that some condensation class between  $X_0$  and  $X_1$  is infinite, i.e. ordered as  $\omega$ . It follows that given any two condensation classes of  $\tau$  ordered as  $\omega$ , either they are consecutive or there is another condensation class ordered as  $\omega$  between them; if no two condensation classes ordered as  $\omega$  are consecutive, it follows that these classes are densely ordered, contradicting the scatteredness of  $\tau$ . ■ Claim

We now claim that any  $A \in [\langle L, < \rangle]^\tau$  can be reduced to an  $A' \in [A]^\tau$  with the following properties:

1. There is at least one  $i$ -canonised  $X \in \mathbf{cc}^\omega(A')$  which is immediately followed by another element of  $\mathbf{cc}^\omega(A')$ ;
2. Let  $\mathcal{X}(A') \subseteq \mathbf{cc}^\omega(A')$  denote the set of all  $X$  as described in condition 1: then amongst those  $X = \{x_n : n \in \omega\} \in \mathcal{X}(A')$  with  $\delta_i(X) := \delta_i(x_0, x_1)$  minimal, there is a leftmost such  $X$ .

Condition 1 is easy to satisfy, as given any  $X \in \mathbf{cc}^\omega(A)$  immediately followed by another element of  $\mathbf{cc}^\omega(A)$  we can simply reduce  $X$  to some  $X' \in [X]^\omega$  which is  $i$ -canonised. For condition 2, suppose there are multiple  $Z \in \mathcal{X}(A)$  all taking the same value of  $\delta_i(Z)$ . Pick one,  $X$ , say, and reduce to  $A' \in [A]^\tau$  by, for each of these  $Z < X$ , removing  $z_0$ ; then for any such  $Z$ ,  $Z \setminus \{z_0\} \in \mathbf{cc}^\omega(A')$  has  $\delta_i(Z') = \delta_i(z_1, z_2)$ , which is strictly greater than  $\delta_i(z_0, z_1) = \delta_i(Z)$  as  $Z$  is  $i$ -canonised by assumption.

Let us write  $\mathcal{E} \subseteq [\langle L, < \rangle]^\tau$  for the set of all  $A \in [\langle L, < \rangle]^\tau$  satisfying conditions 1 and 2 above, and write  $\delta_i^{\min}(A) := \min\{\delta_i(X) : X \in \mathcal{X}(A)\}$ . Our dense set  $\mathcal{D}$  in this case is the set of all those  $A \in \mathcal{E}$  with the additional property that

$$\delta_i^{\min}(A) = \min\{\delta_i^{\min}(A') : A' \in [A]^\tau \cap \mathcal{E}\};$$

the distinguished interval of such an  $A$  is the condensation class  $X$  described in condition 2. ¬Case 2

We are now ready to define our colouring  $c : \mathcal{D} \rightarrow 2$ . For  $A \in \mathcal{D}$ , let us write  $\text{di}(A)$  for the distinguished interval of  $A$ , and write  $A^+ := \{x \in A : x > \text{di}(A)\}$  for the final segment of  $A$  above its distinguished interval. In each case,  $\text{di}(A)$  is immediately followed by (at least) two consecutive elements of  $A$ ; write these as  $x_A < y_A$ . Now, if  $A \in \mathcal{D}$  is such that  $i$  is neither order-preserving nor order-reversing on  $A^+$ , set

$$c(A) := \begin{cases} 0 & \text{if } i(x_A) <_{\text{lex}} i(y_A); \\ 1 & \text{if } i(x_A) >_{\text{lex}} i(y_A). \end{cases}$$

If instead  $i$  is order-preserving or order-reversing on  $A^+$ , set

$$c(A) = C(i \restriction A^+),$$

where  $C$  is the colouring defined in [GSW26, Theorem 2], which simultaneously witnesses that  $\langle \alpha 2, <_{\text{lex}} \rangle \not\vdash (\varphi)^\varphi$  for all well-orderable scattered  $\varphi$  with  $\omega\omega^* \leq \varphi$  or  $\omega^*\omega \leq \varphi$ .

**Claim 15.4.** Let  $A \in \mathcal{D}$  and let  $x < y \in A^+$ . Then there exists  $A' \in [A]^\tau \cap \mathcal{D}$  with  $x = x_{A'}$ ,  $y = y_{A'}$ .

*Proof of claim.* Let us write  $\tau^+$  for the order type of  $A^+$ ; we find our required  $A'$  by reducing  $A^+$  to some  $B \in [A^+]^{\tau^+}$  and leaving  $A \setminus A^+$  untouched. Observe that any  $A' \in [A]^\tau$  of this form is necessarily still in  $\mathcal{D}$ , and has the same distinguished interval as  $A$ : In case 1,  $B$  may contain new almost  $i$ -canonised intervals ordered as  $\omega^*$ ,  $Z$ , say; by our assumption that  $d_2^{\min}(A)$  was already minimal, however, it must be the case that  $d_2(Z) \geq d_2(\text{di}(A))$  for any such  $Z$ , and so  $\text{di}(A)$  is still the leftmost almost  $i$ -canonised element of  $[A']_c^{\omega^*}$  with  $d_2(\text{di}(A))$  minimal; it follows that  $A' \in \mathcal{D}$  and  $\text{di}(A') = \text{di}(A)$ . Similarly, in case 2, any  $Z \in \mathcal{X}(B)$  must have  $\delta_i(Z) \geq \delta_i(\text{di}(A))$ , and so  $\text{di}(A') = \text{di}(A)$ .

We find our desired  $B$  like so. Since  $\tau$  is strictly indecomposable to the right,  $A^+ \cap (y, \rightarrow)$  contains isomorphic copies of any final segment of  $A$ ; let  $B \in [A^+]^{\tau^+}$  be of the form  $B = \{x, y\} \cup B'$ , where  $B' \subseteq A^+ \cap (y, \rightarrow)$ . Then set  $A' := (A \setminus A^+) \cup B$ . Now  $\text{di}(A') = \text{di}(A)$ , and since  $\text{di}(A)$  is immediately followed by  $x$  and  $y$  in  $A'$ , we have that  $x_{A'} = x$  and  $y_{A'} = y$ . ■<sub>Claim</sub>

We now show that no  $A \in \mathcal{D}$  can be homogeneous for  $c$ . Let  $A \in \mathcal{D}$  be arbitrary. If for some  $A' \in [A]^\tau \cap \mathcal{D}$ ,  $i$  is either order-preserving or order-reversing on  $A'^+$ , reduce to  $A'$ ; then since  $i \restriction A'^+$  is not homogeneous for  $C$ , we can find some  $B \subseteq i \restriction A'^+$  with the same order type under  $<_{\text{lex}}$  but with  $C(B) \neq C(i \restriction A'^+)$ ; then  $A'' := (A \setminus A') \cup i^{-1}(B)$  has  $c(A'') = C(B) \neq c(A')$ , and so  $A$  is not homogeneous for  $c$ .

So suppose instead that no such  $A'$  exists. In particular, since  $i$  is neither order-preserving nor order-reversing on  $A^+$ , we can find  $x_0 < y_0$ ,  $x_1 < y_1$  in  $A^+$  such that  $i(x_0) <_{\text{lex}} i(y_0)$  and  $i(x_1) >_{\text{lex}} i(y_1)$ . Then by Claim 15.4, we can find  $A_0, A_1 \in [A]^\tau \cap \mathcal{D}$  with  $x_{A_0} = x_0$ ,  $y_{A_0} = y_0$ ,  $x_{A_1} = x_1$ , and  $y_{A_1} = y_1$ ; then  $c(A_0) = 0$  and  $c(A_1) = 1$ , and so once again  $A$  is not homogeneous for  $c$ . □

*Proof of Theorem 13.* Let  $\tau$  be a well-orderable scattered order type with  $\omega\omega^* \leq \tau$  or  $\omega^*\omega \leq \tau$ . We appeal to the following result of Laver:

**Fact 13.1.** ([Lav71]) Any scattered linear order can be written as a finite sum of indecomposable scattered linear orders.<sup>9</sup>

We can therefore decompose  $\tau$  as a sum  $\tau = \tau_0 + \tau_1 + \dots + \tau_n$ , where each  $\tau_k$  is indecomposable; fix some such decomposition. Given a copy of  $\tau$ , there are potentially many different ways of decomposing it as an interval ordered as  $\tau_0$  followed by an interval ordered as  $\tau_1$ , etc. Whichever of  $\omega\omega^*$  or  $\omega^*\omega$  embeds in  $\tau$  must embed in at least one  $\tau_k$ ; fix some  $m$  such that  $\omega\omega^* \leq \tau_m$  or  $\omega^*\omega \leq \tau_m$ . We define our colouring on a dense subset  $\mathcal{D} \subseteq [L, <]^\tau$  such that for any  $A \in \mathcal{D}$ ,

<sup>9</sup>See the footnote after Fact 15.1.

it is not necessarily possible to pick out an interval of  $A$  ordered as  $\tau_m$ , but it is possible to pick out a final segment of an interval of  $A$  ordered as  $\tau_m$ .

Since  $\tau_m$  is scattered and indecomposable, it is either strictly indecomposable to the right or to the left, and since it is scattered and infinite, it must have infinite condensation classes. Wlog  $\tau_m$  is strictly indecomposable to the right. For  $A \in [\langle L, < \rangle]^\tau$ ,  $X \in [A]_c^\omega \cup [A]_c^{\omega*}$ , we associate each  $X$  with the quantity  $d_0(X) = (\delta_i(x_0, x_1), \delta_i(x_1, x_2))$ . Now define  $\mathcal{E} \subseteq [\langle L, < \rangle]^\tau$  to be the set of all  $A \in [\langle L, < \rangle]^\tau$  with the following properties:

1. There is some  $i$ -canonised  $X \in [A]_c^\omega \cup [A]_c^{\omega*}$  such that for some decomposition  $A = A_0 \cup A_1 \cup \dots \cup A_n$  of  $A$  into consecutive intervals ordered as  $\tau_0, \tau_1, \dots, \tau_n$ , respectively,  $X \in A_m$ ;
2. Amongst all  $X$  as described in condition 1 with  $d_0(X)$  minimal, there is a leftmost such  $X$ .

Observe that  $\mathcal{E}$  is dense: given any  $A \in [\langle L, < \rangle]^\tau$ , we may decompose it into pieces  $A_0 \cup A_1 \cup \dots \cup A_n$  and then  $i$ -canonise some  $\omega$ - or  $\omega^*$ -interval of  $A_m$ , yielding condition 1; given multiple such  $i$ -canonised  $Z \in [A]_c^\omega \cup [A]_c^{\omega*}$  with  $d_0(Z)$  minimal, we may fix some  $X$  and remove  $z_1$  for each such  $Z < X$  to attain condition 2.

Now, for any  $A \in \mathcal{E}$ , let  $X = X(A)$  be the interval guaranteed by condition 2; then there is *some* decomposition of  $A$  into consecutive intervals ordered as  $\tau_k$  such that  $X$  is in the piece ordered as  $\tau_m$ . We cannot in general determine this decomposition from  $X$  alone, nor even the interval ordered as  $\tau_m$ , but since  $\tau_m$  is strictly indecomposable to the right,  $X$  is immediately followed in  $A$  by a final segment of a copy of  $\tau_m$  (in particular,  $X$  is not cofinal in  $\tau_m$ , as  $\omega < \tau_m$  and  $\tau_m$  is strictly indecomposable to the right), and this final segment is uniquely determined; it is the set  $\text{fi}(A) := \{a \in A : a > X \text{ and } \tau_m \not\leq [x_0, a]_A\}$ .

We therefore have a canonical way of picking out an interval of  $A$  ordered as some type equimorphic with  $\tau_m$ ,  $\text{fi}(A)$ ; we would now like to simply apply the colouring from the proof of Lemma 14 to  $\text{fi}(A)$ . In order for this to work, we need to reduce further to another dense subset of  $\mathcal{E}$ : for  $A \in \mathcal{E}$ , write  $d_0^{\min}(A) := d_0(X(A))$ , and let  $\mathcal{D} \subseteq \mathcal{E}$  be the set of those  $A \in \mathcal{E}$  such that  $d_0^{\min}(A)$  is minimal amongst  $\{d_0^{\min}(A') : A' \in [A]^\tau \cap \mathcal{E}\}$ . Then for any  $A \in \mathcal{D}$  and any  $B \in [\text{fi}(A)]^{\text{otp fi}(A)}$ ,  $A' := (A \setminus \text{fi}(A)) \cup B \in [A]^\tau \cap \mathcal{D}$  and  $\text{fi}(A') = B$ . It follows that the colouring  $c' : \mathcal{D} \rightarrow 2$  given by, for  $A \in \mathcal{D}$ ,

$$c'(A) = c(\text{fi}(A)),$$

where  $c$  is the colouring defined in the proof of Lemma 14, has no homogeneous set.  $\square$

Our final result in this section shows that the graph-theoretic IEPR shown consistent in Proposition 5 yields a failure not just of  $\text{KWP}_1$  but also of  $\mathcal{O}$ .

**Proposition 16.** *Let  $G$  be a graph which is linearly orderable, i.e. whose vertex set is linearly orderable. Then*

$$G \not\rightarrow (R)^R,$$

where  $R$  is the random graph.

*Proof.* Our proof uses some ideas from [BCD00, Theorem 2], adapted to the choiceless setting. The key property of the random graph for the purposes of this proof is the following.

**Fact 16.1.** The random graph  $R$  has the *pigeonhole property*, i.e. whenever (the vertex set of)  $R$  is partitioned into finitely many disjoint pieces  $R = R_0 \cup R_1 \cup \dots \cup R_n$ , there is some  $m$  such that  $R_m \cong R$ .

See e.g. [Ca13, Proposition 3] for a proof; Choice is not used. Let  $G$  be a graph with  $[G]^R \neq \emptyset$ , and let  $<$  be a linear ordering on  $G$ . We build a colouring of  $c : [G]^R \rightarrow 2$  with no homogeneous set in the following way. The colour of  $A \in [G]^R$  will be determined by its order-theoretic properties under the ordering  $<$ , and those of its subcopies of  $R$ . If  $A$  is such that there are  $A' \in [A]^R$  with a  $<$ -minimal element and  $A'' \in [A]^R$  with no  $<$ -minimal element, set

$$c(A) := \begin{cases} 0 & \text{if } A \text{ has a minimal element;} \\ 1 & \text{if } A \text{ has no minimal element.} \end{cases}$$

If this is not the case, but  $A$  is such that some  $A' \in [A]^R$  has a maximal element and some  $A'' \in [A]^R$  has no maximal element, set

$$c(A) := \begin{cases} 0 & \text{if } A \text{ has a maximal element;} \\ 1 & \text{if } A \text{ has no maximal element.} \end{cases}$$

It is clear from just this partial definition of  $c$  that for  $A \in [G]^R$  to be homogeneous for  $c$ , we can assume (by reducing if necessary) that either every  $A' \in [A]^R$  has a minimal element or no  $A' \in [A]^R$  has a minimal element, and similarly for maximal elements.

We show first that in order for  $A$  to be homogeneous, it cannot be the case that every  $A' \in [A]^R$  has neither a minimal element nor a maximal element. Let  $a < b \in A$ ; then define  $A_0, A_1 \subseteq A$  by

$$\begin{aligned} A_0 &:= \{a\} \cup \{x \in A : b < x\}; \\ A_1 &:= A \setminus A_0. \end{aligned}$$

Then  $A_0$  has minimal element  $a$  and  $A_1$  has maximal element  $b$ . By the pigeonhole property, at least one of  $A_0, A_1$  is isomorphic to  $R$ , and so we obtain some  $A_i \in [A]^R$  which has either a minimal element or a maximal element.

If  $A$  is homogeneous it therefore must either be the case that every element of  $[A]^R$  has a minimal element or that every element of  $[A]^R$  has a maximal element.

**Claim 16.2.** Let  $A$  be such that every element of  $[A]^R$  has a minimal element. Then there exists  $S \in [A]^R$  which is well-ordered by  $<$ .

*Proof of claim.* Let  $S$  be the maximal well-ordered  $<$ -initial segment of  $A$ ; we claim that  $S \in [A]^R$ . To see this, observe that  $A \setminus S$  has no minimal element; if  $a = \min A \setminus S$ , then  $(\leftarrow, a] \cap A = S \cup \{a\}$  is well-ordered by  $<$ , as  $S$  is well-ordered by  $<$  and  $a > S$ , so  $a \in S$ , a contradiction. By the pigeonhole principle, one of  $S$  or  $A \setminus S$  is in  $[A]^R$ ; since all elements of  $[A]^R$  have a minimal element, this must be  $S$ . ■ Claim

Similarly, if  $A$  is such that every element of  $[A]^R$  has a maximal element, there is some  $S \in [A]^R$  which is anti-well-ordered by  $<$ .

We now complete the definition of our colouring. If  $A \in [G]^R$  is well-ordered by  $<$ , set

$$c(A) := \begin{cases} 0 & \text{if } a_0 E a_1; \\ 1 & \text{if } \neg a_0 E a_1, \end{cases}$$

where  $a_0, a_1$  are the first two elements of  $A$  under  $<$ ; similarly, if  $A$  is anti-well-ordered by  $<$ , set

$$c(A) := \begin{cases} 0 & \text{if } a_0 E a_1; \\ 1 & \text{if } \neg a_0 E a_1, \end{cases}$$

where  $a_0, a_1$  are the last two elements of  $A$  under  $<$ . In all remaining undefined cases, the colour of  $A$  does not matter; set  $c(A) = 0$ .

We claim that this colouring has no homogeneous set. Let  $A \in [G]^R$ . For  $A$  to be homogeneous for  $c$ , it must be the case either that every element of  $[A]^R$  has a minimal element or that every element of  $[A]^R$  has a maximal element; wlog every element of  $[A]^R$  has a minimal element. Then we may assume, by reducing further if necessary, that  $A$  is well-ordered by  $<$ , and that  $\text{otp}\langle A, < \rangle$  is minimal in  $\{\text{otp}\langle A', < \rangle : A' \in [A]^R\}$ . Then for any  $B \subseteq A$  which has strictly smaller order type than  $A$ , it must be the case that  $A \setminus B \in [A]^R$ . In particular, letting  $x, y \in A$ ,  $x < y$  be such that

$$x E y \iff \neg a_0 E a_1,$$

we see that  $A' := \{x, y\} \cup \{a \in A : a > y\}$  is an element of  $[A]^R$ ; but since  $a'_0 = x$  and  $a'_1 = y$ , we have  $c(A) \neq c(A')$ , and so  $A$  is not homogeneous for  $c$ . □

We remark that exactly the same proof gives a more general result. Let us say that a relation  $r$  of arity  $n$  on a structure  $A$  is *trivialised* by the linear order  $\langle A, < \rangle$  if whenever  $a_0 < a_1 < \dots < a_{n-1}$  and  $a'_0 < a'_1 < \dots < a'_{n-1}$ ,

$$A \models r(a_0, a_1, \dots, a_{n-1}) \iff r(a'_0, a'_1, \dots, a'_{n-1}).$$

The final step of the proof of Proposition 16 simply relies on the fact that the edge relation  $E$  is not trivialised by any linear ordering of  $R$ . As such, we obtain the following generalisation:

**Theorem 17.** *Let  $\mathcal{L}$  be a relational language, let  $A$  be an  $\mathcal{L}$ -structure whose underlying set is linearly orderable and let  $B$  be an  $\mathcal{L}$ -structure with the pigeonhole property with at least one relation which is not trivialised by any linear order on  $B$ . Then*

$$A \not\rightarrow (B)^B.$$

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